Demonstration : $\sigma = \frac{k_f}{k_i} |f|^2$.

On a $dn = \underbrace{\sigma \vec{F}_i \, d\Omega}_{\text{Avant diffusion.}} = \underbrace{\vec{C}\, \vec{j}_d \, dS}_{\text{Après diffusion}}$ ($\vec{F}_d$)

on a $\vec{F}_i = C \vec{j}_i$ et $d\Omega = \frac{dS}{r^2}$

$\Rightarrow \sigma C j_i \frac{dS}{r^2} = C . j_d . dS$

$\Rightarrow \sigma = \frac{j_d}{j_i} . r^2$

$j = \frac{\hbar k_i}{\mu}$

$$\begin{cases} j_d(r) = \frac{\hbar k_f}{\mu} \frac{|f|^2}{r^2} \\ j_d(\theta) \propto \frac{1}{r^3} \\ j_i(\varphi) \propto \frac{1}{r^3} \end{cases} \Big\} \text{negligeable devant } \frac{1}{r^2}$$

Alors $\sigma = \dfrac{\frac{\hbar k_f}{\mu} \frac{|f|^2}{r^2} \times r^2}{\frac{\hbar k_i}{\mu}}$

et on trouve que : $\boxed{\sigma = \frac{k_f}{k_i} |f|^2}$

Calcule Amplitude de Diffusion.

- Les états stationnaires d'énergie E sont de la forme $\psi(r,t) = e^{-\frac{iEt}{\hbar}} \psi(r)$, où $\psi(r)$ vérifie l'équat° :

$$\left(\frac{-\hbar^2}{2m}\Delta + V(r)\right)\psi(r) = E\psi(r)$$

$$\left(\frac{-\hbar^2}{2m}\Delta + V(r) - E\right)\psi(r) = 0$$

$E = \frac{\hbar^2 k^2}{2m}$

$$\Rightarrow \boxed{\left(\Delta + k^2 - U(r)\right)\psi(r) = 0} \quad (1)$$

avec $U(r) = \frac{2m}{\hbar^2} V(r)$

- Nous démontrons l'existance de fonct° d'onde stationnaire ayant de la forme de :

$$V_k^{dif}(r) \underset{r\to\infty}{\simeq} e^{ikz} + f_k(\theta,\varphi)\frac{e^{ikr}}{r}$$

- On écrit l'équat° (1) sous forme.

$$\boxed{(\Delta + k^2)\psi(r) = U(r)\psi(r)} \quad (2)$$

* Supposant que l'on connaisse une fonct° $G(r)$ tel que. $(\Delta + k^2)G(r) = \delta(r)$

avec $G(r)$ fonct° de Green pour l'operateur Δ.

* toute fonct° $\varphi(r)$ vérifiant.

$$\varphi(r) = \varphi_0(r) + \int G|r-r'| . \psi(r') d^3(r') U(r')$$

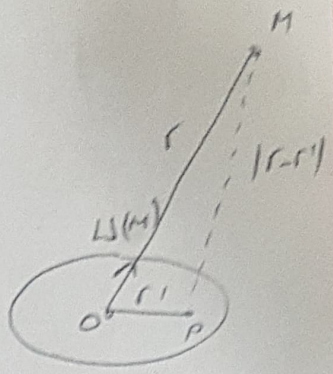
est une solution de l'équat° (2) avec φ_0 est une solut° de l'équat° libre $(\Delta + k^2)\psi(r) = 0$

- en effet, on montre que.

$$G^+|r-r'| = -\frac{1}{4\pi}\frac{e^{ik|r-r'|}}{|r-r'|}$$

- On démontre que $G^{\pm}(r) = \frac{-1}{4\pi} e^{\pm ikr}$

$\begin{cases} G^+ : \text{f de Green sortant} \\ G^- : \text{f de Green entrant.} \end{cases}$



$$|r-r'| \simeq \vec{r} - \vec{r}'\vec{u}$$
$$\simeq r - \vec{u}\vec{r}'$$
$$\simeq r - r'\cos\theta$$

- l'eq integrale de la diffusion est.

$$V_k^{diff}(r) = e^{ikz} + \int G(r-r')\, U(r')\, V_k^{dif}(r')\, d^3r'$$

- le comportement asymtotique de cette éq est.

$$V_k^{diff}(r) = e^{ikz} - \frac{1}{4\pi r} e^{ikr} \int e^{-ik\vec{u}r'} U(r')\, V_k^{diff}(r')\, dr'$$

$$= e^{ikz} - \frac{1}{4\pi r} e^{ikr} \int e^{-ik_f \vec{r}'} U(r')\, V_k^{diff}(r')\, dr'$$

$$V_k(r') = \psi_{k_i}^+(r)$$

on trouve.

$$f_k(\theta,\varphi) = \frac{-1}{4\pi} \int e^{-ik_f r'} U(r')\, \psi_{k_i}^+(r')\, dr'$$

$$\boxed{f_k(\theta,\varphi) = \frac{-m}{2\pi\hbar^2} \int e^{-ik_f r'} V(r')\, \psi_{k_i}^+(r')\, dr'}$$

- Les fonct° d'ondes non relativiste atome hydrogéne dans son etath 1s et 2s :

- Généralement :

$$\varphi_{1s} = \frac{1}{\sqrt{\pi}} \left(\frac{Z}{a_0}\right)^{3/2} e^{\frac{-Z}{a_0} r}$$

$$\varphi_{2s} = \frac{1}{\sqrt{32\pi}} \left(\frac{Z}{a_0}\right)^{3/2} \left(2 - \frac{Zr}{a_0}\right) e^{\frac{-Z}{2a_0} r}$$

Pour l'atome Hydrogéne : $Z = 1$ $a_0 = 1$

$$\varphi_{1s} = \frac{1}{\sqrt{\pi}} e^{-r}$$

$$\varphi_{2s} = \frac{1}{\sqrt{32\pi}} (2 - r) e^{-r/2}$$

$$\frac{1}{r_{01}} = \sum_{l=0}^{\infty} \sum_{m=-l}^{l} \frac{4\pi}{2l+1} \frac{r_<^l}{r_>^{l+1}} Y_{lm}(\hat{\mathbf{r}}_1) Y_{lm}^*(\hat{\mathbf{r}}_0)$$

$$\exp(i\boldsymbol{\Delta}.\mathbf{r}_0) = 4\pi \sum_{l=0}^{\infty} \sum_{m=-l}^{l} i^l J_l(\Delta r_0) Y_{lm}(\hat{\mathbf{r}}_0) Y_{lm}^*(\hat{\boldsymbol{\Delta}})$$

$$R_{1,0}(r) = 2a_0^{-3/2} \exp\left(-\frac{r}{a_0}\right); \quad R_{2,0}(r) = 2(2a_0)^{-3/2} \left(1 - \frac{r}{2a_0}\right) \exp\left(-\frac{r}{2a_0}\right)$$

$$\int_0^{+\infty} dx\ x^n e^{-\alpha x} = \frac{n!}{\alpha^{n+1}}; \quad \int x^3 \exp(-3x/2) J_1(\Delta x) dx == \frac{12\Delta}{((\frac{3}{2})^2 + \Delta^2)^3}$$

$$\psi_{2p_0} = \frac{1}{2^{3/2}\sqrt{3}} r_1 \exp(-\frac{1}{2} r_1) Y_{10}(\hat{r}_1); \quad Y_{10}(\hat{r}_1) = \sqrt{\frac{3}{4\pi}} cos(\theta)$$

Bon courage

$1s\ 1s$

$$A_{1s1s} = -e^{-2r_0}\left(1 + \frac{1}{r_0}\right)$$

$$T_d = -(2\pi^2)^{-1}\ \frac{8 + \Delta^2}{(4+\Delta^2)^2}$$

$$\frac{d\sigma}{d\Omega} = 4\ \frac{(8+\Delta^2)^2}{(4+\Delta^2)^4}$$

$$\sigma = \frac{4\pi}{k^2}\left[\frac{-7k^6 - 18k^4 - 12k^2}{12(k^2+1)^3}\right]$$

$$= \frac{7\pi}{3k^2}.$$

1s 2s

$$\begin{cases} A = \frac{3}{2} r_0^2 e^{-\frac{3}{2} r_0} \\ B = \frac{2}{27} e^{-\frac{3}{2} r_0} \left(4 + 6 r_0 - 9 r_0^2 \right) \end{cases}$$

$$A_{1s2s} = \frac{4}{9\sqrt{2}} e^{-\frac{3}{2} r_0} \cdot \left(\frac{1}{r_0} + \frac{2}{3} \right)$$

$$T_d = (2\pi)^{-3} \frac{16\sqrt{2}\,\pi}{\left(\frac{9}{4} + \Delta^2 \right)^3}$$

$$\sigma = \frac{128}{5 k_i^2} \left(\frac{2}{3} \right)^{10}$$

1S ⟶ 2P

$$\psi_{2p} = \frac{\sqrt{2}}{8\sqrt{\pi}} \cos(\theta_1)\, r_1\, e^{-r_1/2} \quad \Big| \quad Y_{10}(\Delta) = \sqrt{\frac{3}{4\pi}}\cos(\theta_\Delta)$$

$$J_1(\Delta r_1) = \frac{\sin \Delta r_1}{(\Delta r_1)^2} - \frac{\cos \Delta r_1}{\Delta r_1}$$

$$\begin{cases} A = \dfrac{1}{\Delta}\, \dfrac{\frac{27}{4} + 3\Delta^2}{\left(\frac{9}{4} + \Delta^2\right)^3} \\ B = \dfrac{1}{\Delta}\, \dfrac{\frac{27}{4} - 3\Delta^2}{\left(\frac{9}{4} + \Delta^2\right)^3} \end{cases} \Rightarrow T_\alpha = (2\pi)^{-3}\, \frac{24\pi\sqrt{2}}{\Delta\left(\frac{9}{4} + \Delta^2\right)^3}$$

$$\frac{d\sigma}{d\Omega} = \frac{k_f}{k_i}\, \frac{288}{\Delta^2\left(\frac{9}{4} + \Delta^2\right)^6}$$

Programme 1s 1s

```
Programm: elastique 1s1s
ISSE MSIMSLMS
  implicit intiger (i-n)
  implicit real*8 (a-h, o-y)
open(5, file 'SEDNonre1s1s.dat', status='unknown')

    evua = 27,2116
      pi = 4.0*atan(1.0)
       C = 137,036

  do 201    igei = 100, 100, 100
      Ecii = float(igei)/evua
      Eci = float(igei)
  pimr = Sqrt(2.0*Ecii)
  ~~float(pimr*pimr)~~

  do 200 itetaf = -90, 90
     adtf = float(itetaf)
     apipf = adtf*pi/180
  delt1s = 2*pimr*Sin(apipf/2)

  Sed1s = 4.*(delt1s**2+8.)**2/(delt1s**2.+4.)**4
  Seg1s = 7.*pi/3.*Pimr**2
Close(5)
```

Programme 1h 2h

```
Programme inelastique 1h 2h
ISBE MSIMSLMS
   implicit integer (i-n)
  implicit real*8 (a-h,o-y)
Open(5, file 'SEDNonre 1h 2h .dat' status='unknown'
      evisa = 27,2116
       pi = 4.0*atn(1.0)
        c = 137,036

  do 201   igei = 100
       Ecii = float(igei)/evua
       Eci, = float(igei)
     pimr = sqrt(2.0*Ecii)
     pfmr = sqrt(pimr*pimr - 3/4.)
  do 200   itetaf = -90,90
    atdf = float(itetaf)
    apipf = float(itetaf)* pi/180.
  delt2h = pimr**2 + pfmr**2 - 2.*pimr*pfmr*cos(apipf)

Sed2h = (pfmr/pimr)* 128/(9.14+ del2h**2)**6

Seg2h = (128* Pi/(5.*Pimr**2.))*(2/3)**10
Close(5)
```